

The Third Law of Thermodynamics Nobody knows what entropy really is, so in any discussion you will always have to use the word. There are many possible ways of sticking pictures; I will show you two possible picture arrangements. There are many possible ways of sticking pictures; I will show you two possible picture arrangements. Large system has more entropy than the small system. You can also increase the total entropy of a system by adding more particles.

Now imagine you are sticking pictures on your wall A , nearby is my wall B . I ask if you could decorate my wall too, so our total entropy of wall $A + B$ system. When you start moving pictures to my wall, the total entropy of the system increases.

Inductively, we imagine the universe as a giant closed system, with many particles inside; and they are always exchanging energy with each other.

An ice cube is more "organized" than a cup of water, and vice versa. In the middle of winter, water is frozen. Does this phenomenon violate the second law of thermodynamics? Surprisingly, the answer is no. The water is cooled down since it comes in contact with a colder reservoir.

But:

Entropy is closely related to energy, but they are not the same, and we do not have any law of conservation of entropy; so we can increase the total entropy of a system.

So the total entropy of our system (a cup of water and the outside environment) increases, because it gains more than it loses.

And obviously, also by the law:

The temperature itself is not so important than the **difference**. If our system only consists of objects at same temperature, no net change in entropy.

In this section, we develop the connection among three physical quantities: (**heat, energy, entropy**). Let's begin with a review of the first two.

Plot our data into a simple graph: [H]

A simple graph of the entropy. Looking at the table and graph, we find out some interesting things. The total entropy is increasing. Can I guess the slope of a tangent line at point $q_A = 20$ is **steeper** than point $q_B = 10$?

Again, the total entropy is increasing! Can I guess the slope of a tangent line at point $q_A = 20$ is **steeper** than point $q_B = 10$?

Substituting our values yields:

Here is a question, why is ΔS_B negative, but it still has positive slope? The answer is pretty simple, our graph perspective is different.

According to the zeroth law of thermodynamics, the temperature of solid A and B are equal if and only if they are in equilibrium.

A subtle observation is that a decrease in q_A corresponds to an increase in q_B :

This equation makes perfect sense to our intuition, look at previous calculation: the slope of S_A graph is much steeper than the slope of S_B graph.

To ensure the second law of thermodynamics, as you can see here, the entropy of solid A increases **faster** than the rate of entropy change of solid B .

Apparently, we can see that: